

1 Computational model

Our model consists of three neuronal populations (C_1 , C_2 , and O), artificially separated for clarity rather than representing strictly segregated neural populations. To generate activity from the animal's simulated trajectories, we employ a linear-non-linear Poisson approach: simulating firing rate dynamics via a continuous attractor neural network (CANN) and generating discrete spikes from a Poisson distribution parameterized by the computed rates.

For any neuron i in the cell population p , we model its membrane potential $U_i^{(p)}(t)$ via firing rate dynamics:

$$\tau \frac{d}{dt} U_i^{(p)}(t) = -U_i^{(p)}(t) + I_{\text{in},i}^{(p)}(t), \quad (1)$$

where τ is the membrane time constant and $I_{\text{in},i}^{(p)}(t)$ is the total net input current to the cell. The firing rate $r_i^{(p)}(t)$ is modeled as a nonlinear function of the membrane potential:

$$r_i^{(p)}(t) = g^{(p)} \frac{[U_i^{(p)}(t)]_+^2}{1 + k^{(p)} \sum_j [U_j^{(p)}(t)]_+^2}, \quad (2)$$

where $g^{(p)} > 0$ is a gain factor, $[\cdot]_+$ rectifies the content taking the maximum value between it and zero and $k^{(p)}$ determines the global inhibition strength that stabilizes activity and prevents runaway excitation. Spike counts for a given time bin of length Δt are then obtained through a pseudo-random Poisson process:

$$P(n \text{ spikes}) = \frac{(r_i(t)\Delta t)^n e^{-r_i(t)\Delta t}}{n!} \quad (3)$$

Neurons in each population p are assigned a preferred spatial phase $\Phi_i^{(p)} = (\Phi_i^x, \Phi_i^y)$ on a rhombic domain, corresponding to a hexagonal lattice of firing fields in physical space. Conjunctive cells are additionally assigned a preferred head direction $\theta_i^{(p)}$. Synaptic connections depend on the phase distance on a twisted toroidal manifold, $\|\cdot\|_{\text{Torus}}$, which accounts for the periodic boundaries of the grid lattice. In general, connectivity matrices (both recurrent and feedforward) are built according to the distance between neurons within this embedding:

$$W_{ij}^{p_2 \rightarrow p_1} = W_0^{p_2 \rightarrow p_1} \left[\exp \left(-\frac{\left\| \Phi_i^{(p_1)} - \Phi_j^{(p_2)} - \delta^{p_2 \rightarrow p_1} \hat{e}(\theta_j^{(p_2)}) \right\|_{\text{Torus}}^2}{2\sigma_T^2} \right) - T \right]_+ \quad (4)$$

where $\hat{e}(\theta_j)$ is the unit vector in the preferred head direction of cell j in population p , δ a shift in the toroidal manifold, $W_0 > 0$ represents the maximal

synaptic strength, $T > 0$ determines the interaction radius and σ_T the width of synaptic extent in the toroidal manifold.

The first layer comprises primary conjunctive grid-by-head-direction cells. These cells integrate self-motion and spatial inputs. Their input current $I_{in,i}^{C_1}$ is driven by bottom-up spatial visual inputs (I_{vis}), local head direction tuning (I_{hd}), and internal recurrent collateral connections (W^{rec}):

$$I_{in,i}^{C_1}(t) = I_{hd}(t) \left(I_i^{vis}(t) + \sum_j W_{ij}^{rec} r_j^{C_1}(t) \right), \quad (5)$$

$$\text{with } I_{hd}(t) = \exp \left(-\frac{\|\Theta(t) - \theta_i^{C_1}\|_{ang}^2}{2\sigma_{HD}^2} \right)$$

where $\|\cdot\|_{ang} = \min\{|\cdot|, 2\pi - |\cdot|\}$ is the angular distance. For sake of simplicity, we model the head direction input as a multiplicative current, an common approach in these type of models (SOURCE). Besides, we also exclude the speed inputs and simulate the system in simulated trajectories of constant speed to avoid calibration burdens. Apart from this, recurrent weights will be also influenced on the head direction preference, making weights less strong for very different angular preferences. Applying the matrix prescription base on distance on the twisted torus W_{ij} we define

$$W_{ij}^{rec} = W_{ij} \exp \left(-\frac{\|\theta_i^{C_1} - \theta_j^{C_2}\|_{ang}^2}{2\sigma_{HD}^2} \right) \quad (6)$$

The second layer consists of intermediate conjunctive cells driven by feedforward projections from C_1 and modulated by vestibular signals. The net input current is:

$$I_{in,i}^{C_2}(t) = \gamma_i(t) \sum_j W_{ij}^{C_1 \rightarrow C_2} r_j^{C_1}(t) \quad (7)$$

The feedforward weights $W_{ij}^{1 \rightarrow 2}$ share the identical shifted Gaussian topology as the recurrent weights, projecting a predicted spatial phase. The vestibular gain $A_{vest,i}(t)$ incorporates slope-dependent modulation:

$$A_{vest,i}(t) = 1 + \beta_{vest} \sin(\alpha) \exp \left(-\frac{(\theta_S - \theta_i)^2}{2\sigma_S^2} \right)$$

where β_{vest} controls the strength of the modulation associated with slope angle α in the inclination direction θ_S , and σ_S dictates the directional tuning. While this gain does not affect path integration in the primary layer, it enhances the feedforward contribution of cells with an uphill-pointing preferred direction.

Omnidirectional Grid Layer (O)

The final layer comprises omnidirectional grid cells. These cells lack intrinsic directional preference and receive convergent feedforward input from C_2 :

$$I_{in,i}^O(t) = \sum_j W_{ij}^{2 \rightarrow o} r_j^{C_2}(t)$$

The weights $W_{ij}^{2 \rightarrow o}$ map the directionally tuned predictive activity back into a purely spatial representation. Because C_2 activity is selectively amplified by the vestibular gain, the resulting center-of-mass in the O layer shifts upward on inclined surfaces, reproducing the experimentally observed non-directional shift.